

OSLAB PRACTICAL-1 PROGRAMS & OUTPUT

Date: 22-08-26

Program-1 & OUTPUT:

```
1w 1 1 1 sainath sainath 755 Aug 22 04:01 program2.c
sainath@Sainath:~/OSLAB/practical_5$ cat program1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>
#include <string.h>

int main() {
    int fd[2];
    char buffer[100];
    char data[] = "Hello from Producer";

    pipe(fd);

    pid_t pid = fork();

    if (pid > 0) {
        // Parent - Producer
        close(fd[0]);

        clock_t start = clock();

        write(fd[1], data, strlen(data) + 1);

        close(fd[1]);

        wait(NULL);

        clock_t end = clock();

        printf("Producer sent: %s\n", data);
        printf("Communication time: %f seconds\n",
            (double)(end - start) / CLOCKS_PER_SEC);
    }
    else {
        // Child - Consumer
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Consumer received: %s\n", buffer);

        close(fd[0]);
    }

    return 0;
}
sainath@Sainath:~/OSLAB/practical_5$ ./program1
Consumer received: Hello from Producer
Producer sent: Hello from Producer
Communication time: 0.000033 seconds
```


Program-2 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_5$ cat program2.c
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int fd[2];

    pipe(fd);

    pid_t p1 = fork();

    if (p1 == 0) {
        // Child 1: ls -l
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);

        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }

    pid_t p2 = fork();

    if (p2 == 0) {
        // Child 2: grep ".c"
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);

        close(fd[0]);

        execlp("grep", "grep", ".c", NULL);

        perror("execlp");
        exit(1);
    }

    // Parent
    close(fd[0]);
    close(fd[1]);

    waitpid(p1, NULL, 0);
    waitpid(p2, NULL, 0);

    return 0;
}
```

```
sainath@Sainath:~/OSLAB/practical_5$ ./program2
```

```
-rw-r--r-- 1 sainath sainath 823 Aug 22 03:59 program1.c
-rw-r--r-- 1 sainath sainath 739 Aug 22 04:01 program2.c
```

```
sainath@Sainath:~/OSLAB/practical_5$ ls -l | grep ".c"
-rw-r--r-- 1 sainath sainath 823 Aug 22 03:59 program1.c
-rw-r--r-- 1 sainath sainath 739 Aug 22 04:01 program2.c
```